



GRFormer: Grouped Residual Self-Attention for Lightweight Single Image Super-Resolution

Yuzhen Li

Tsinghua University
Shenzhen, Guangdong, China
liyuzhen22@mails.tsinghua.edu.cn

Yuxin Cao

National University of Singapore
Singapore
e1374346@u.nus.edu

Zehang Deng*

Swinburne University of Technology
Melbourne, Australia
zehangdeng@swin.edu.au

Lihua Liu

University of International Business and Economics
Beijing, China
202102027@uibe.edu.cn

Abstract

Previous works have shown that reducing parameter overhead and computations for transformer-based single image super-resolution (SISR) models (e.g., SwinIR) usually leads to a reduction of performance. In this paper, we present GRFormer, an efficient and lightweight method, which not only reduces the parameter overhead and computations, but also greatly improves performance. The core of GRFormer is Grouped Residual Self-Attention (GRSA), which is specifically oriented towards two fundamental components. Firstly, it introduces a novel grouped residual layer (GRL) to replace the Query, Key, Value (QKV) linear layer in self-attention, aimed at efficiently reducing parameter overhead, computations, and performance loss at the same time. Secondly, it integrates a compact Exponential-Space Relative Position Bias (ES-RPB) as a substitute for the original relative position bias to improve the ability to represent position information while further minimizing the parameter count. Extensive experimental results demonstrate that GRFormer outperforms state-of-the-art transformer-based methods for $\times 2$, $\times 3$ and $\times 4$ SISR tasks, notably outperforming SOTA by a maximum PSNR of 0.23dB when trained on the DIV2K dataset, while reducing the number of parameter and MACs by about **60%** and **49%** in only self-attention module respectively. We hope that our simple and effective method that can easily applied to SR models based on window-division self-attention can serve as a useful tool for further research in image super-resolution. The code is available at <https://github.com/sisrformer/GRFormer>.

CCS Concepts

• Computing methodologies → Visual inspection.

Keywords

Lightweight image super-resolution; Self-Attention; Parameter overhead

*Corresponding Author



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1 Introduction

Single Image Super-Resolution (SISR) aims to enhance image resolution by reconstructing a high-resolution image from a low-resolution counterpart. With the development of CNN-based and transformer-based SR models, one achievement after another has been achieved for single image super-resolution tasks. For whether CNN-based models or transformer-based ones, it is an easy way to improve performance by increasing the number of the network layers and feature dimension, accompanied by the increase of parameters and calculations. A straight question should be: Is it possible to improve the performance while reducing number of the parameters and calculations for transformer-based SR models? Motivated by this question, we conduct in-depth research into self-attention and present three sub-questions concerning self-attention:

- **RQ1.** Is there any redundancy within self-attention?
- **RQ2.** Can the expressive power of self-attention be further improved?
- **RQ3.** Is there a better alternative to relative position bias for representing the position information?

The research into RQ1. Self-attention [16] has achieved great performance in the fields of text, image, and video since it was published in 2017. However, despite the effectiveness of self-attention mechanisms, they are often criticized for their extensive parameter count and computational demands. Existing work [2] has observed redundancy in self-attention layers, but its solutions focus mainly on how to reduce the size of the attention window [17]. Through empirical analysis, we explore the interaction among the varying parameter counts, MACs, and the performance of the self-attention module, as depicted in Fig. 1. This analysis reveals significant potential for optimization in both parameters and computational efficiency within the SwinIR's self-attention mechanism. Inspired by these findings, we propose a novel grouping scheme of Q, K, V linear layer, aimed at reducing both the parameter overhead and computational complexity.

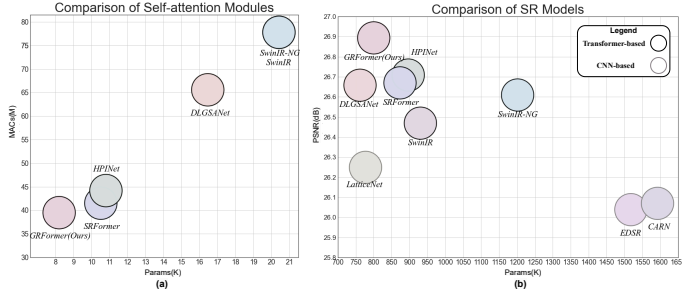


Figure 1: (a) shows the comparisons of self-attention of recent transformer-based SR models in terms of multiply-accumulate operations (MACs) and parameters. (b) shows SISR comparisons of recent SR models ($\times 4$) in terms of PSNR on Urban100, network parameters. Our model (GRFormer) outperforms the SOTA model ($\times 4$) by 0.19dB in PSNR score while having comparably low network parameters and MACs.

The research into RQ2. In the field of single image super-resolution, previous transformer-based approaches [8, 18] have primarily utilized residual learning [5] at the outer layers of self-attention mechanisms. This methodology has been effective in mitigating network degradation and improving the expressive capabilities of deep networks. Given the success of residual learning in enhancing network performance, an intriguing question arises: could the integration of residual connections within the Query, Key, and Value (QKV) linear layers of self-attention mechanisms further augment their expressive power? Motivated by this consideration, our work explores the incorporation of residual connections directly into the QKV linear layers, aiming to enhance their representational efficacy in deep neural architectures.

The research into RQ3. When self-attention [16] is proposed, it uses position embedding to provide position information of words in text. After self-attention is applied to computer vision, relative position bias (RPB) is found to be more suitable for representing relative position information, which is important for models in computer vision. However, RPB has four fatal flaws: First, if we assume the shape of the window in self-attention is $H \times W$, the number of parameters occupied by RPB is $(2H-1) \times (2W-1)$, which is a huge parameter overhead. Second, the original RPB in SwinIR[8] sets a position parameter for each position within the window, which is not only the redundancy of the parameters, but also it is easy to be interfered by noises during training. Specifically, many subtle fluctuations can be found in the figures of Fig. 2 (a). Third, during training, each parameter of RPB is trained independently, ignoring the relative relationship between the weights of different positions. Fourth, RPB fails to clearly reflect intuition: For reconstructing an image, near pixels tend to be more important than far pixels. Motivated by this, we designed an Exponential-Space Relative Position Bias (ES-RPB) to replace RPB.

First and foremost, to solve the **RQ1**, we propose a grouping scheme for the QKV linear layer. Furthermore, to solve the **RQ2** and compensate for performance loss from the grouping scheme, we add residuals for the QKV linear layer. Lastly, to solve **RQ3**, we propose

Exponential-Space Relative Position Bias (ES-RPB). The three methods above constitute the two fundamental components of Grouped Residual Self-Attention (GRSA). Given the proposed GRSA, we design a lightweight network for SR, termed GRFormer. We evaluate our GRFormer on five widely-used datasets. Benefiting from the proposed GRSA, our GRFormer achieves significant performance improvements on almost five benchmark datasets. Notably, trained on DIV2K dataset[15] for $\times 2$ SR task, our GRFormer achieves a 33.17 PSNR score on the challenging Urban100 dataset[6]. The result is much higher than recent SwinIR-light[8](32.76) and the SOTA lightweight SR model (32.94). This improvement is consistently observed across $\times 3$ and $\times 4$ tasks. Comprehensive experiments show that GRFormer not only outperforms previous lightweight SISR models [8, 10, 18], but also reduces the parameter count by about 20% in total model architecture, compared with SwinIR [8] (1000k parameters) with the same hyperparameter. To sum up, our contributions can be summarized as follows:

- We propose a novel Grouped Residual Self-Attention (GRSA) for lightweight image super-resolution, which can not only reduce the parameter count but also enhance the performance in an easy-to-understand way for SR tasks. In addition, our proposed GRSA module can seamlessly replace the self-attention module and its variants in other transformer-based SR models, simultaneously reducing the number of parameter by about 60% and the number of MACs by about 49% in only self-attention module.
- Based on GRSA, we construct a novel transformer-based SR network, termed GRFormer. Our GRFormer achieves state-of-the-art performance in lightweight image SR, and outperforms previous lightweight SISR networks by a large margin in most of the five benchmark datasets.

2 Related Works

2.1 CNN-Based Image Super-Resolution

A lot of CNN-based SR models [1, 9, 13] have emerged since SRCNN [4] introduces CNN-based deep learning method for image SR. By removing unnecessary modules in conventional residual networks and using a new multi-scale deep super-resolution system, EDSR [9] achieves significant performance improvement. By implementing a cascading mechanism upon a residual network, CARN [1] reduces the amount of computation. To further reduce the amount of computation, LatticeNet [13] applies two butterfly structures to combine two residual blocks.

2.2 Transformer-based Image Super-Resolution

Since Swin Transformer [12] introduces hierarchical architecture and shifted windowing scheme, the feasibility of transformer application in the field of computer vision has been greatly improved. In order to introduce transformer into the field of image super-resolution, SwinIR[8] is proposed, which has been baseline model for transformer-based SR models. However, for some lightweight scenarios, the amount of parameters and calculations in SwinIR is still too large and its relative position bias lacks a certain organization.

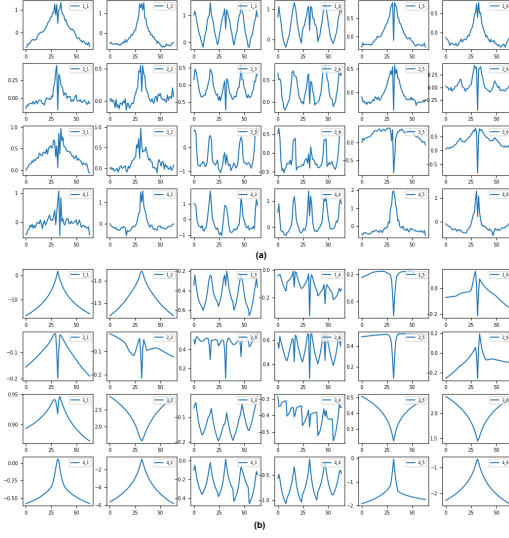


Figure 2: Comparison between RPB and ES-RPB. The subfigure (a) and (b) showcase the relative position bias (RPB) from the SwinIR model and an GRFormer model where RPB are replaced to ES-RPB, respectively. The subfigure at i_{th} row and j_{th} column corresponds to the relative position bias of i_{th} GRSAB Group and j_{th} GRSAB in the network. These figures specifically highlight the horizontal evolution of the relative position bias values. The x-axis extends from 0 to 62, while the y-axis corresponds to the data taken at the 7th point on the x-axis from an RPB matrix of size 15×63 .

2.3 Relative position bias

When self-attention is proposed to solve problems in the field of natural language processing (NLP), absolute position embedding is designed to supplement position information of words in text. After self-attention is applied into image super-resolution, relative position bias enjoys more popularity than the absolute position embedding, because relative position bias can provide the relative position information, which is intrinsically more suitable for image super-resolution. However, there are some problems concerning relative position bias that cannot be ignored, such as parameter redundancy, weak ability to resist interference during training and so on. Afterwards, in order to tackle resolution gaps between pre-training of large vision models, SwinIR-v2 [11] proposes a log-spaced continuous position bias method to effectively transfer large-scale models pre-trained using low-resolution images to downstream tasks with high-resolution inputs. Although the log-spaced continuous position bias is designed to solve the problem of resolution difference of pre-trained large-scale models, it is very inspiring for the design of relative position bias of transformer-based SR models.

3 Method

3.1 Overall Architecture

The overall architecture of our GRFormer is shown in Fig. 3, consisting of three parts: shallow feature extraction, deep feature extraction, and image reconstruction. Given the LR input $I_{LR} \in$

$R^{H \times W \times C_{in}}$, we first use a 3×3 convolution L_{SF} to transform the low-resolution image I_{LR} to shallow feature $X_0 \in R^{H \times W \times C}$ as

$$X_0 = L_{SF}(I_{LR}) \quad (1)$$

where C_{in} and C is the channel number of LR input and shallow feature. This convolution layer simply converts the input from image space into high-dimensional feature space. Then, we use N grouped residual self-attention block groups L_{GRSABG} and a 3×3 convolution layer L_{conv} at the end to extract the deep feature $I_{DF} \in R^{H \times W \times C}$. The process can be expressed as

$$\begin{aligned} X_i &= L_{GRSABG_i}(X_{i-1}), \\ I_{DF} &= L_{conv}(X_N) + X_0 \end{aligned} \quad (2)$$

In GRSAB Group, given X_i as input, we use M grouped residual self-attention block L_{GRSAB} to get $X_{i,M}$. Then we use a 3×3 convolution layer L_{conv} to get X_{i+1} . The process can be expressed as

$$\begin{aligned} X_{i,0} &= X_i, \\ X_{i,j} &= L_{GRSAB_j}(X_{i,j-1}), \\ X_{i+1} &= L_{conv}(X_{i,M}) + X_{i,M} \end{aligned} \quad (3)$$

Finally, we use a 3×3 convolution layer L_{conv} to get better feature aggregation, and aggregate shallow and deep features to reconstruct HR image $I_{HR} \in R^{H \times W \times C_{out}}$ as

$$I_{HR} = L_{shuffle}(L_{conv}(I_{DF}) + I_{SF}), \quad (4)$$

where C_{out} is the channel number of the high-resolution image and $L_{shuffle}$ is a PixelShuffle [14] module.

3.2 Grouped Residual Self-Attention Block

The Grouped Residual Self-Attention Block (GRSAB) mainly consists of two core components: Grouped Residual Self-Attention (GRSA), Feed Forward Network module (FFN). Given the input of GRSAB as $I_{in} \in R^{H \times W \times C}$, we first use a grouped residual self-attention module L_{GRSA} to learn the global relationships of pixels in a window. After L_{GRSA} , we use a LayerNorm module to normalize the feature, because the normalized features can eliminate gradient vanishing and make the training stable. The process can be expressed as

$$I_{GRSA} = Norm(L_{GRSA}(I_{in})) + I_{in} \quad (5)$$

where I_{GRSA} is the output of GRSA module. Then we use a feed forward network to transform the I_{GRSA} to another feature space and a LayerNorm module to normalize the feature as

$$I_{out} = Norm(L_{FFN}(I_{GRSA})) + I_{GRSA} \quad (6)$$

3.3 Grouped Residual Self-Attention

To reduce the number of parameters as well as calculations and enhance the performance, we introduce the Grouped Residual Self-Attention (GRSA), which incorporates two novel and compact components: grouped residual layer (GRL) and exponential-space relative position (ES-RPB). GRL consists of two parts: residuals for QKV linear layer to transform the feature space of self-attention from the linear space to residual space, grouping scheme for QKV linear layer to reduce the parameters and calculations. By reducing the noise interference during training and making self-attention sensitive to pixel distance, ES-RPB improves the expression ability of position information. The proposed GRSA aggregates features

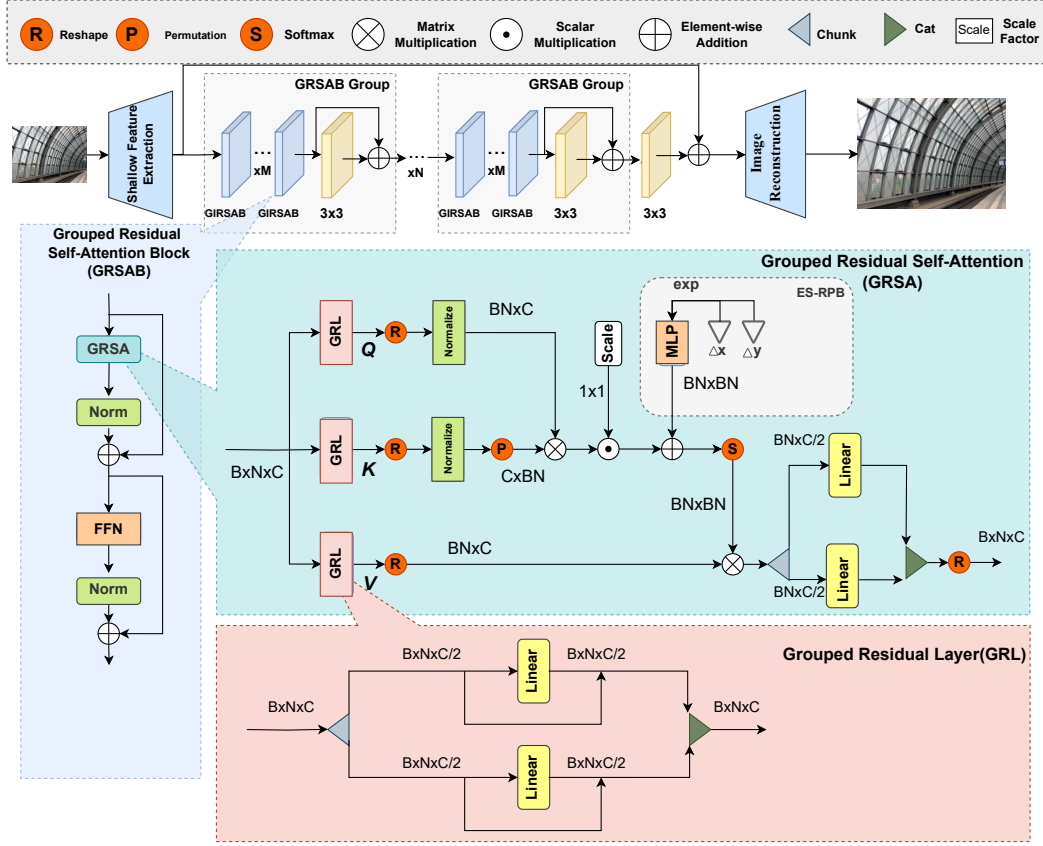


Figure 3: Network architecture of the proposed GRFormer. It mainly consists of a shallow feature extraction module, several grouped residual self-attention block groups (GRSAB Group) to learn deep feature mapping in an efficient and effective way, and a high-resolution image reconstruction module.

of pixels globally in the window. Specifically, given input of GRSA as $X \in R^{H \times W \times C}$, we uses L_{GRL_Q} , L_{GRL_K} , L_{GRL_V} to get Q, K, V.

$$\begin{aligned} Q &= L_{GRL_Q}(X), \\ K &= L_{GRL_K}(X), \\ V &= L_{GRL_V}(X) \end{aligned} \quad (7)$$

where L_{GRL_Q} , L_{GRL_K} , L_{GRL_V} are the GRL module corresponding to Q, K, V. Then we normalize Q and K and calculate the matrix product of Q and K to get the similarity of Q and K. We multiply QK^T by a trainable self-attention scaling factor λ , add B_{ES-RPB} and perform a Softmax operation. Next, we calculate the matrix product of the QK^T after Softmax and V. If multi-head self-attention is applied, we will use a grouped linear layer L_{proj} as a projection at the last to map the multi-head to one head. The formulation can be written as:

$$\begin{aligned} GRSA &= L_{proj}(\text{Softmax}(\lambda \cdot \text{Normalize}(Q) \\ &\quad * \text{Normalize}(K)^T + B_{ES-RPB})V) \end{aligned} \quad (8)$$

3.4 Grouped Residual Linear.

As a substitute for the QKV linear layer, the Grouped Residual Linear (GRL) is one of the core modules of GRSA, with the objective

of reducing the amount of parameters and calculations and basically maintaining the feature learning ability. By integrating concepts of *grouping* and *residuals* into the self-attention mechanism, we enhance its structural efficiency and functional effectiveness.

- **Grouping scheme of QKV linear layer in self-attention.**

A lot of works, such as [18], show that there is redundancy in generation of Q, K, V and matrix product of Q, K. To reduce the redundancy, we apply the idea of grouping. Given the input as X, we divide X into two equal parts in the channel dimension and then use two independent linear layers to get Q, K, V respectively. Grouping scheme of Q, K, V will not significantly reduce the interaction of pixel features, because the matrix multiplication of Q and K in self-attention will offset these shortcomings to some extent, which is discussed in section 3.6.

- **Residuals of the Q, K, V linear layers.** Residuals allows training to be performed in the residual space, which makes the network find the optimal solution in the residual space. So we add it into the QKV linear layer to transform the training space of QKV linear layer from linear space to residual space in order to enhance the feature learning ability of QKV linear layer.

Specifically, we assume the input of GRL module is $X \in R^{H \times W \times C}$. We first divide X in channel dimension into two parts X_{in_1} , X_{in_2} .

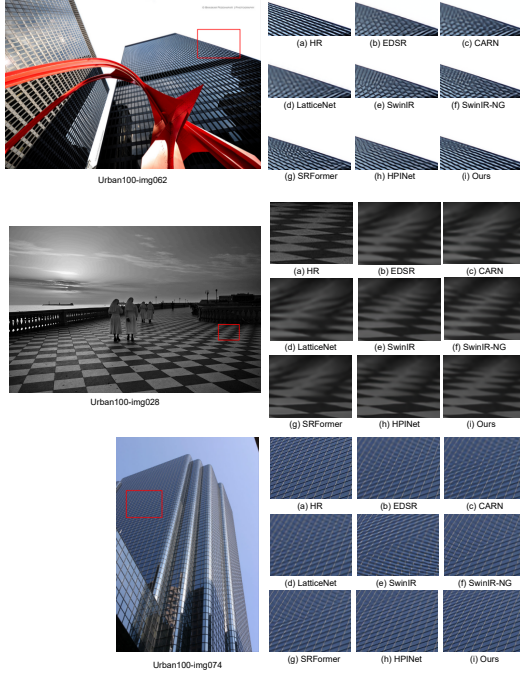


Figure 4: Qualitative comparison with recent state-of-the-art lightweight image SR methods on the $\times 4$ SR task.

Then, by using residuals, each uses a linear layer to get X_{out_1}, X_{out_2} . Finally, we merge X_{out_1}, X_{out_2} in channel dimension to get the output X_{out} at the last as

$$\begin{aligned} X_{in_1}, X_{in_2} &= X, \\ X_{out_1} &= L_{Linear_1}(X_{in_1}) + X_{in_1}, \\ X_{out_2} &= L_{Linear_2}(X_{in_2}) + X_{in_2}, \\ X_{out} &= X_{out_1}, X_{out_2}, \end{aligned} \quad (9)$$

where L_{Linear_1} and L_{Linear_2} are linear layers, $X_{in_1}, X_{in_2}, X_{out_1}, X_{out_2} \in R^{H \times W \times \frac{C}{2}}$.

3.5 Exponential-Space Relative Position Bias

To solve the four disadvantages of original RPB mentioned in RQ3, we propose the Exponential-Space Relative Position Bias (ES-RPB). We design an exponential mapping for original absolute position coordinates to forcibly add pixel distance sensitive rules to original RPB, which makes it give more weight to nearby pixels. What's more, we use a tiny multilayer perceptron (MLP) to obtain the mapping of all absolute position coordinates, which reduces the impact of noise during training and makes it easier to be trained. Specifically, we transform the abscissa ΔX and the ordinate ΔY in absolute position coordinate matrix from linear space to exponential space, and then we use a tiny MLP to get B_{ES-RPB}

$$\begin{aligned} \Delta \hat{X} &= \text{sign}(\Delta X) * (1 - \exp(-|\alpha * \Delta X|)), \\ \Delta \hat{Y} &= \text{sign}(\Delta Y) * (1 - \exp(-|\beta * \Delta Y|)), \\ B_{ES-RPB} &= \text{MLP}(\Delta \hat{X}, \Delta \hat{Y}) \end{aligned} \quad (10)$$

where α and β is trainable distance-sensitive factors to control the sensitivity to distances between reference pixel and the others in the same window and the symbol $*$ represents the multiplication of each position in the matrix. MLP consists of two linear layers and an activation layer sandwiched between them.

3.6 Explanation of the effectiveness of GRL

GRL consists of two parts: a grouping scheme and a residual structure. Grouping scheme is proposed to efficiently reduce the parameters and calculations without severe performance degradation, and residual structure makes the network find the optimal solution in the residual space, thereby reducing the difficulty of feature learning. The explanation is as follows:

- **Explanation of the effectiveness of grouping scheme.** First, we analyse the effectiveness in reduction of parameters and MACs. We assume that the number of input features is N . Then, both the parameters and the MACs occupied by the linear layer are N^2 . But, if we group the input features into halves and use two linear layers with the number of input features of $\frac{N}{2}$ to process the two parts of input features, respectively, both parameters and MACs occupied by the two linear layers are only $\frac{N^2}{2}$. Obviously, the grouping scheme can halve the number of parameters and MACs.

Second, we analyse the effectiveness analysis in preventing severe performance degradation. A possible concern about the grouping scheme is that grouping features will result in a lack of aggregation of the two groups of features, leading to performance degradation. We will analyze below that, at least for Q, K matrices, this worry is unnecessary. As shown in the self-attention formula: $\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$, Q and K matrices are only used to perform matrix product and get QK^T matrix. During the matrix product of Q and K, even if we employ the grouping scheme to divide the input features into two groups and perform feature aggregation individually, the subsequent matrix product of Q and K will deeply aggregate the two groups of features. The process can be shown as follows. Specifically, given input of self-attention as $X \in R^{N \times C}$, where N is the number of pixels in a window and C is number of input features, we cut up X into 2×2 blocks as

$$\begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix} = X, \quad (11)$$

where $X_{11}, X_{12}, X_{21}, X_{22} \in R^{\frac{N}{2} \times \frac{C}{2}}$. By packing parameter matrices of two grouped linear layers with input features of $\frac{C}{2}$ into one parameter matrix, we can assume the parameter matrices of Q, K projections as $\begin{pmatrix} M_{Q_1} \\ M_{Q_2} \end{pmatrix}, \begin{pmatrix} M_{K_1} \\ M_{K_2} \end{pmatrix}$ respectively, where $M_{Q_1}, M_{Q_2}, M_{K_1}, M_{K_2} \in R^{\frac{C}{2} \times \frac{C}{2}}$. Then we can perform matrix product of X and the parameter matrices of Q, K projections respectively as

$$\begin{aligned} Q, K &= \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix} * \begin{pmatrix} M_{Q_1 \vee K_1} \\ M_{Q_2 \vee K_2} \end{pmatrix} \\ &= \begin{pmatrix} X_{11} * M_{Q_1 \vee K_1} & X_{12} * M_{Q_2 \vee K_2} \\ X_{21} * M_{Q_1 \vee K_1} & X_{22} * M_{Q_2 \vee K_2} \end{pmatrix}, \end{aligned} \quad (12)$$

Table 1: Quantitative evaluations of the lightweight GRFormer against state-of-the-art methods on commonly used benchmark datasets. Best and second best results are marked in red and blue colors. #Params means the number of the network parameters. #MACs denotes the number of the MACs which are calculated on images with an upscaled spatial resolution of 1280×720 pixels. #Weighted-Avg means the weighted average PSNR and SSIM on five benchmark datasets. #Weighted-Avg = $\sum_{i=1}^5 M_i \times \text{Score}_i$, where M is the number of images in the dataset and Score is the corresponding PSNR or SSIM score.

Scale	Method	Year	#Params/(K)	#MACs/(G)	Set5	Set14	B100	Urban100	Manga109	#Weighted-Avg
×2	EDSR-baseline[9]	CVPRW2017	1370	316.3	37.99/0.9604	33.57/0.9175	32.16/0.8994	31.98/0.9272	38.54/0.9769	34.37/0.9353
	CARN [1]	ECCV2018	1592	222.8	37.76/0.9590	33.52/0.9166	32.09/0.8978	31.92/0.9256	38.36/0.9765	34.27/0.9342
	LatticeNet [13]	ECCV2020	756	169.7	38.15/0.9610	33.78/0.9193	32.25/0.9005	32.43/0.9302	-/-	-/-
	SwinIR-light [8]	ICCV2021	910	207.5	38.14/0.9611	33.86/0.9206	32.31/0.9012	32.76/0.9340	39.12/0.9783	34.87/0.9386
	SwinIR-NG [3]	CVPR2023	1181	210.7	38.17/0.9612	33.94/0.9205	32.31/0.9013	32.78/0.9340	39.20/0.9781	34.90/0.9385
	DLGSANet-light [7]	ICCV2023	745	169.4	38.20/0.9612	33.89/0.9203	32.30/0.9012	32.94/0.9355	39.29/0.9780	34.98/0.9389
	SRFormer-light [18]	ICCV2023	853	198.6	38.23/0.9613	33.94/0.9209	32.36/0.9019	32.91/0.9353	39.28/0.9785	34.98/0.9393
	HPINet-M [10]	AAAI2023	783	213.1	38.12/-	33.94/-	32.31/-	32.85/-	39.08/-	34.88/-
	GRFormer (Ours)	-	781	198.4	38.22/0.9614	34.01/0.9214	32.35/0.9018	33.17/0.9375	39.30/0.9785	35.07/0.9399
×3	EDSR-baseline [9]	CVPRW2017	1555	160.1	34.37/0.9270	30.28/0.8417	29.09/0.8052	28.15/0.8527	33.45/0.9439	30.38/0.8692
	CARN [1]	ECCV2018	1592	118.6	34.29/0.9255	30.29/0.8407	29.06/0.8034	28.06/0.8493	33.50/0.9440	30.36/0.8676
	LatticeNet [13]	ECCV2020	765	76.2	34.53/0.9281	30.39/0.8424	29.15/0.8059	28.33/0.8538	-/-	-/-
	SwinIR-light [8]	ICCV2021	918	94.2	34.62/0.9289	30.54/0.8463	29.20/0.8082	28.66/0.8624	33.98/0.9478	30.76/0.8746
	SwinIR-NG [3]	CVPR2023	1190	95.67	34.64/0.9293	30.58/0.8471	29.24/0.8090	28.75/0.8639	34.22/0.9488	30.89/0.8757
	DLGSANet-light [7]	ICCV2023	752	75.8	34.70/0.9295	30.58/0.8465	29.24/0.8089	28.83/0.8653	34.16/0.9483	30.89/0.8759
	SRFormer-light [18]	ICCV2023	861	88.3	34.67/0.9296	30.57/0.8469	29.26/0.8099	28.81/0.8655	34.19/0.9489	30.90/0.8764
	HPINet-M [10]	AAAI2023	924	110.6	34.70/-	30.63/-	29.26/-	28.93/-	34.21/-	30.95/-
	GRFormer (Ours)	-	789	93.5	34.67/0.9293	30.64/0.8481	29.27/0.8100	29.07/0.8702	34.35/0.9494	31.04/0.8781
×4	EDSR-baseline [9]	CVPRW2017	1518	114.2	32.09/0.8938	28.58/0.7813	27.57/0.7357	26.04/0.7849	30.35/0.9067	28.14/0.8119
	CARN [1]	ECCV2018	1592	90.88	32.13/0.8937	28.60/0.7806	27.58/0.7349	26.07/0.7837	30.47/0.9084	28.19/0.8118
	LatticeNet [13]	ECCV2020	777	43.6	32.30/0.8962	28.68/0.7830	27.62/0.7367	26.25/0.7873	-/-	-/-
	SwinIR-light [8]	ICCV2021	930	54.18	32.44/0.8976	28.77/0.7858	27.69/0.7406	26.47/0.7980	30.92/0.9151	28.51/0.8204
	SwinIR-NG [3]	CVPR2023	1201	55	32.44/0.8980	28.83/0.7870	27.73/0.7418	26.61/0.8010	31.09/0.9161	28.62/0.8221
	DLGSANet-light [7]	ICCV2023	761	43.2	32.54/0.8993	28.84/0.7871	27.73/0.7415	26.66/0.8033	31.13/0.9161	28.65/0.8227
	SRFormer-light [18]	ICCV2023	873	53	32.51/0.8988	28.82/0.7872	27.73/0.7422	26.67/0.8032	31.17/0.9165	28.67/0.8230
	HPINet-M [10]	AAAI2023	896	81.1	32.60/-	28.87/-	27.73/-	26.71/-	31.19/-	28.69/-
	GRFormer (Ours)	-	800	50.8	32.58/0.8994	28.88/0.7886	27.75/0.7431	26.90/0.8097	31.31/0.9183	28.80/0.8260

where $\vee$ represents the logical symbol "or". Then, we perform the matrix product of Q and K as follows:

$$QK^T = \begin{pmatrix} O_{11} & O_{12} \\ O_{21} & O_{22} \end{pmatrix}, \quad (13)$$

then, each element in matrix of QK^T is calculated:

$$\begin{aligned} O_{11} &= X_{11}M_{Q_1}M_{K_1}^T X_{11}^T + X_{12}M_{Q_2}M_{K_2}^T X_{12}^T, \\ O_{12} &= X_{11}M_{Q_1}M_{K_1}^T X_{21}^T + X_{12}M_{Q_2}M_{K_2}^T X_{22}^T, \\ O_{21} &= X_{21}M_{Q_1}M_{K_1}^T X_{11}^T + X_{22}M_{Q_2}M_{K_2}^T X_{12}^T, \\ O_{22} &= X_{21}M_{Q_1}M_{K_1}^T X_{21}^T + X_{22}M_{Q_2}M_{K_2}^T X_{22}^T \end{aligned} \quad (14)$$

It can be seen from Equation 14, in the matrix product of Q and K of the self-attention mechanism, both X_{11} and X_{12} are used to perform a series of matrix products to get O_{11} , O_{12} , and so do X_{21} and X_{22} to get O_{21} , O_{22} , which aggregates two groups of features. Therefore, the matrix product of Q and K will offset the weak aggregation of input features brought by the use of grouping scheme. In other words, our grouping scheme won't lead to lack of aggregation of the two groups of features and thereby won't lead to severe performance degradation.

- **Explanation of the effectiveness of residual structure for the QKV linear layer.** As shown in Formula 8, self-attention involves multiple matrix products, which makes it difficult for the network to learn the optimal parameters. Given the input of self-attention as X and the optimal Q, K, V as $\hat{Q}$, $\hat{K}$, $\hat{V}$, if we add the residual structure for QKV linear layer, the network only needs to optimize $\hat{Q}-X$, $\hat{K}-X$, $\hat{V}-X$ instead of $\hat{Q}$, $\hat{K}$, $\hat{V}$. The residual structure transforms the learning space of QKV linear

layer from linear space to residual space, enhancing the feature learning ability of QKV linear layer.

3.7 Explanation of the effectiveness of ES-RPB

We mainly analyze the effectiveness of ES-RPB from the perspective of parameter reduction and performance improvement for ES-RPB.

- **Explanation of parameter reduction for ES-RPB.** We assume that the size of the window of self-attention is $W \times H$ and the self-attention head is 1. The number of parameters occupied by the original RPB is $(2 \times W - 1) \times (2 \times H - 1)$. Specifically, the window size is usually set to 16×16 , so the parameter count is 961 in this case. In contrast, if we suppose that the number of features of the hidden layer in MLP is C_{hidden} , the amount of parameters occupied by ES-RPB is $3C_{hidden}$. Specifically, C_{hidden} is set to 128 in our GRFormer, so the number of parameters occupied by ES-RPB in GRFormer is 384, which is much less than 961. What has to be noted is that, when the height and width of window grow simultaneously at a linear rate, the parameters occupied by original RPB will grow squarely, while the parameters occupied by ES-RPB will not grow.
- **Explanation of performance improvement for ES-RPB.** The ES-RPB mechanism within GRFormer improves performance through three key strategies. Firstly, instead of training positional parameters directly, which can be noise-sensitive, we utilize a tiny MLP to get the B_{ES-RPB} . This approach minimizes the noise impact on positional parameters during training. Secondly, the mechanism enhances interaction of parameters representing relative position information by training them through this tiny MLP

rather than in isolation. Thirdly, ES-RPB introduces a distance-sensitive design. It employs an exponential function to map the absolute positional coordinates $(\Delta X, \Delta Y)$ from linear space to exponential space, resulting in $\Delta \hat{X}$ and $\Delta \hat{Y}$. This transformation ensures that positions closer to the reference pixel exhibit more significant changes, aligning with the principle that nearer pixels should attract more attention.

4 Experiments

In this section, we conduct experiments on the lightweight image SR tasks, compare our GRFormer with existing state-of-the-art methods, and do ablation analysis of the proposed method.

4.1 Experimental Setup

Datasets and Evaluation. For training, we use DIV2K (Agustsson and Timofte 2017), the same as the comparison models, to train our GRFormer. It includes 800 training images and 100 validation images, mainly concerning human, animals, plants, buildings, etc. For testing, we use five public SR benchmark datasets: Set5 (Bevilacqua et al. 2012), Set14 (Zeyde, Elad, and Protter 2010), B100 (Martin et al. 2001), Urban100 (Huang, Singh, and Ahuja 2015) and Manga109 (Matsui et al. 2017) to evaluate model. The experimental results are evaluated in terms of two objective criteria: peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM), which are both calculated on the Y channel from the YCbCr space.

Implementation Details. We set the GRSAB Group number, GRSAB number of a GRSAB Group, feature number, and attention head number, window size to 4, 6, 60, 3, 8×32 , respectively. The training low-resolution patch size we use is 64×64 . When training, we randomly rotate the images by 0° , 90° , 180° , 270° and randomly flip images horizontally for data augmentation. We adopt the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.99$ to train the model for 600k iterations. The learning rate is initialized as 2×10^{-4} and halves on {250000, 400000, 510000, 540000}-th iterations. We use L1 loss to train the model. The whole process is implemented by Pytorch on NVIDIA GeForce RTX 3080 GPUs.

4.2 Comparisons with State-of-the-arts

We compare our GRFormer with commonly used lightweight SR models for upscaling factor $\times 2$, $\times 3$, $\times 4$, including EDSR [9], CARN [1], LatticeNet [13], SwinIR [8], SwinIR-NG [3], HPINet [10], DLGSANet [7], SRFormer [18]. We compare the parameters, calculations as well as performance on five commonly used SR benchmark datasets (Set5, Set14, B100, Urban100, Manga109). The comparison results are grouped for $\times 2$, $\times 3$, $\times 4$ upscaling factor.

Quantitative Comparison Table 1 shows quantitative comparisons in terms of PSNR and SSIM on five benchmark datasets. As shown in Table 1, our GRFormer achieves the best PSNR score and SSIM score for $\times 2$, $\times 3$, $\times 4$ task on Set14, Urban100, Manga109 and weighted average of the five benchmark datasets, and the PSNR score and SSIM score achieved by our GRFormer is either quite close or superior to that of SOTA model on Set5 and B100. What's more, GRFormer outperforms SOTA model on Urban100 and Manga109 by a large margin. It is worth noting that, our GRFormer outperforms SwinIR-light by a maximum PSNR of 0.42dB and SOTA by a maximum PSNR of 0.23dB, which is a significant improvement for

image SR. Furthermore, our GRFormer outperforms SOTA model by about 0.1dB on the weighted average of the five benchmark datasets for $\times 2$, $\times 3$, $\times 4$ task. Although our GRFormer achieves great performance, the parameters and MACs of GRFormer is relatively low. As shown in Fig. 1 (a), compared with the self-attention of other transformer-based SR models, our GRSA has the smallest number of parameters and calculations.

Qualitative Comparison We further show visual examples of common used methods under scaling factor $\times 4$. As shown in Fig. 4, we use three images reconstructed by EDSR, CARN, LatticeNet, SwinIR, SwinIR-NG, SRFormer, HPINet and GRFormer to make qualitative comparisons.

We qualitatively compare models on Urban100-img062, img028, and img074. For img062, EDSR, CARN, LatticeNet, SwinIR, SwinIR-NG, and SRFormer show severe texture and color distortions, especially in the lower right corner, while GRFormer has the smallest distortion and closest color to HR. For img028, models like EDSR, CARN, LatticeNet, and SRFormer exhibit significant blurring and deformation, whereas GRFormer achieves the best reconstruction. For img074, only GRFormer accurately reconstructs the clear window frame direction without distortion, highlighting its superior detail preservation.

4.3 Ablation Analysis

We conduct ablation experiments to study the effect of Grouped Residual Layer (GRL) and Exponential-Space Relative Position Bias (ES-RPB). Ablation experiments are trained on DF2K and evaluated on the Set5, Set14, B100, Urban100, Manga109 datasets. PSNR and SSIM are adopted to evaluate the perceptual quality of recovered images. We also adopt parameters and MACs on images with an upscaled spatial resolution of 1280×720 pixels to evaluate the complexity. The results are shown in Table 2.

Effectiveness of GRL As shown in Table 2, model ① achieves a 47% reduction in #SA-Params, a 17% reduction in #Params, a 43% reduction in #SA-MACs, and a 17% reduction in #MACs compared to model ③, primarily due to optimizations in self-attention. Despite these reductions, GRL maintains performance, demonstrating its effectiveness. Further experiments with models ④ and ⑤ reveal:

- Effectiveness of grouping scheme. Model ① significantly reduces parameters and computations compared to model ④, with minor performance degradation mitigated by the residual structure in the QKV layer.
- Effectiveness of residual structure. Model ④, retaining the same parameters as model ③, significantly improves performance, especially on Urban100 and Manga109, by 0.07dB and 0.08dB, respectively.

Effectiveness of ES-RPB The core feature of ES-RPB is the ability to represent the pixel position information. To highlight the contribution of our ES-RPB, we replace the ES-RPB of model ① with RPB in SwinIR to get model ②. As shown in Table 2, compared with model ②, model ① improves the performance on five benchmark datasets in a large margin. Specifically, model ① outperforms model ② on Urban100 by 0.44dB PSNR score, which is a notable boost in lightweight image super-resolution. To further understand the reason of improvement brought by ES-RPB, we draw the three dimensional view of both ES-RPB in model ① and

Table 2: Effect of the GRFormer on SISR. #SA-Params and #SA-MACs mean the parameters and MACs in our GRSA respectively. #Params and #MACs mean the parameters and MACs in our GRFormer. The ablation experiments are trained on DF2K for $\times 4$ SR task and tested on benchmark datasets (Set5, Set14, B100, Urban100, Manga109) to get PSNR and SSIM.

Method	GRL		ES-RPB	#SA-Params	#Params	#SA-MACs	#MACs	Set5	Set14	B100	Urban100	Manga109
	Group	Residuals										
①	✓	✓	✓	8.2K	810K	39.5M	50.8G	32.59/0.8999	28.92/0.7890	27.77/0.7435	26.97/0.8110	31.47/0.9195
②	✓	✓	✓	10.3K	850K	38.9M	50.8G	32.46/0.8980	28.81/0.7865	27.72/0.7414	26.53/0.7998	31.07/0.9150
③			✓	15.4K	973K	69M	61.4G	32.60/0.9000	28.93/0.7892	27.77/0.7437	27.00/0.8118	31.48/0.9197
④		✓	✓	15.4K	973K	69M	61.4G	32.63/0.9002	28.95/0.7897	27.78/0.7441	27.07/0.8139	31.56/0.9206
⑤	✓		✓	8.2K	810K	39.5M	50.8G	32.58/0.8996	28.91/0.7888	27.76/0.7434	26.97/0.8116	31.43/0.9191
⑥				20.4K	1023K	77.8M	64.8G	32.53/0.8988	28.85/0.7874	27.74/0.7424	26.64/0.8030	31.17/0.9165

Table 3: Quantitative evaluations for $\times 4$ task. GRFormer-8 $\times$ 32, GRFormer-16 $\times$ 16 means GRFormer that uses window size of 8 $\times$ 32, 16 $\times$ 16 respectively.

Method	#Params/(K)	#MACs/(G)	Set5	Set14	B100	Urban100	Manga109	#Weighted-Avg
GRFormer-8 $\times$ 32	800	50.8	32.58/0.8994	28.88/0.7886	27.75/0.7431	26.90/0.8097	31.31/0.9183	28.80/0.8260
GRFormer-16 $\times$ 16	800	50.8	32.53/0.8994	28.89/0.7883	27.75/0.7431	26.92/0.8102	31.34/0.9187	28.81/0.8262

Table 4: Classical SR model comparisons ($\times 4$ SR). We report PSNR on Urban100 and Manga109. All models are trained on DF2K training set.

Method	window size	#Params/(M)	#MACs/(G)	Urban100	Manga109
SwinIR	8 $\times$ 8	11.9	747	27.45	32.03
SRFormer	16 $\times$ 16	10.52	683	27.53	32.07
GRFormer	16 $\times$ 16	9.5	641	27.7	32.16

Table 5: Evaluations of running time (/ms) on NVIDIA GeForce RTX 3080 GPUs.

Method	$\times 2$	$\times 3$	$\times 4$
SwinIR-light	183	125	109
SwinIR-NG	248	185	160
GRFormer-light	205	167	142

RPB in model ②. As shown in Fig.2, we can see that, the curve of ES-RPB in model ① is roughly similar to that of RPB in model ②, because both ES-RPB and RPB represent the relative position information. As shown in Fig.2, after comparing the curves of RPB and ES-RPB, we can find two differences: First, minor fluctuations on the curve of ES-RPB are much less than that of RPB, which means that most of noise interference is removed. Secondly, we can see from the curve of RPB that RPB is overfitted, which will lead to poor generalization ability. In contrast, our ES-RPB use simpler structure to improve generalization ability. It means that RPB learns a lot of content that is not universally applicable, which affects its generalization ability to represent relative position information.

5 Comparison of window size

As shown in Table 3, GRFormer-8 $\times$ 32 outperforms GRFormer-16 $\times$ 16 by 0.05dB in Set5, while GRFormer-16 $\times$ 16 outperforms GRFormer-8 $\times$ 32 by 0.03dB in Manga109. Additionally, compared with Table 1, our GRFormer-16 $\times$ 16 still achieves state-of-the-art results. Therefore, the choice between window sizes 8 $\times$ 32 and 16 $\times$ 16 does not significantly impact the model’s overall performance.

6 Classical Image Super-Resolution

For the classical image SR task, we compare our GRFormer with two state-of-the-art Transformer-based methods: SwinIR, and SRFormer. Our GRFormer, SRFormer, and SwinIR are all 10M-scale

models. As shown in Table 4, for $\times 4$ task, our GRFormer outperforms SwinIR by 0.25dB and outperforms SRFormer by 0.17dB in Urban100. Therefore, our GRFormer still outperforms the state-of-the-art Transformer-based methods in classical Image Super-Resolution.

7 Running time

As shown in Table 5, the running times for the SwinIR-light, SwinIR-NG, and GRFormer-light methods are 109 ms, 160 ms, and 142 ms, respectively, for the $\times 4$ task. These times were measured using images with an upscaled spatial resolution of 256 $\times$ 256 pixels on NVIDIA GeForce RTX 3080 GPUs. Although GRFormer-light exhibits a slightly higher running time compared to SwinIR-light, it still offers a competitive performance with significantly lower running times than SwinIR-NG. Furthermore, GRFormer-light strikes a balance between efficiency and effectiveness, ensuring satisfactory performance while maintaining manageable computation times, thus making it a viable option for practical applications where both speed and quality are critical.

8 Conclusion

In this paper, we propose GRSA, an efficient self-attention mechanism composed of two components: GRL, which significantly reduces parameters and calculations, and ES-RPB, which efficiently and effectively represents distance-sensitive relative position information. The GRL module employs a grouping scheme to minimize redundancy in parameters and calculations with minimal performance loss and uses a residual structure to enhance feature learning for the QKV linear layer. Based on GRSA, we introduce GRFormer, a simple yet effective model for lightweight single image super-resolution. GRFormer leverages GRL and ES-RPB to significantly reduce parameters and MACs while enhancing PSNR and SSIM performance. Experimental results demonstrate GRFormer’s superior performance over previous state-of-the-art lightweight SR models on benchmark datasets, particularly Urban100 and Manga109. We hope GRSA will be a valuable tool for future SR model design research.

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